

# FungalMorphoSpace: A reaction-diffusion simulator for exploring fungal hymenophore morphogenesis

Mario Ahumada Durán<sup>a</sup>

<sup>a</sup>*Independent Researcher,*

---

## Abstract

FungalMorphoSpace is an open-source Python package for simulating and quantifying fungal hymenophore patterns using Gierer–Meinhardt reaction-diffusion systems. The software implements a forward-Euler solver with CFL-adaptive time-stepping, periodic boundary conditions, and robust morphometric analysis including dual-metric wavelength estimation (FFT radial spectrum and autocorrelation, both with quality control), Euler characteristic computation, and connected-component spacing statistics. It ships with empirically calibrated parameter sets for three polypore species—*Fomes fomentarius*, *Lentinus brumalis*, and *Polyporus squamosus*—spanning an  $8\times$  wavelength range (33 px to 235 px, 261  $\mu\text{m}$  to 2043  $\mu\text{m}$ ). Key features include temporal rescaling control ( $T_{\text{target}}$ ) for cross-species comparability, a contract-enforced output schema for reproducibility, sensitivity analysis tools, and an optional three-node (immobile-node) module with a linear-stability utility that lets users reach the same morphospace at equal diffusion—making explicit that pattern scale is governed by a reaction–diffusion length rather than by an extreme diffusion ratio. Parameters are calibrated empirically, so FungalMorphoSpace is a hypothesis-generation and morphospace-exploration tool—not an *ab initio* predictor—for researchers in fungal developmental biology, computational morphogenesis, and bio-inspired materials design.

**Keywords:** Reaction-diffusion, Turing patterns, Fungal morphogenesis, Hymenophore, Pattern formation, Gierer–Meinhardt, Polypores, Computational biology

---

## 1. Motivation and significance

The spore-bearing surfaces of polypore fungi—their hymenophores—display a remarkable diversity of spatial patterns: regular hexagonal pores in *Fomes fomentarius*, fine-grained pores in *Lentinus brumalis*, and coarse pores in *Polyporus squamosus*. Despite apparent morphological differences, Kuhar et al. [1] proposed that these structures represent topological variants of a single self-organizing mechanism governed by reaction-diffusion (RD) dynamics, as originally formulated by Turing [2].

---

Email address: mahumadad@gmail.com (Mario Ahumada Durán)

Testing this hypothesis computationally requires tools that can (i) generate patterns across diverse parameter regimes, (ii) quantify morphological features objectively, and (iii) compare simulated outputs with empirical measurements. We emphasize at the outset what such a tool can and cannot establish: by fitting parameters to observed spacings it can show *consistency*—that the morphologies are reproducible within the model—but it cannot show that fungi *employ* this mechanism. That stronger claim would require deriving parameters from independent biophysical measurements and then predicting the spacing (Sec. 4). FungalMorphoSpace is built for the former, exploratory role. While general-purpose RD simulators exist, none provides an integrated pipeline tailored to fungal morphogenesis that includes species-specific calibration, topological analysis, and reproducible output contracts.

FungalMorphoSpace addresses this gap by providing:

- A modular Gierer–Meinhardt simulator with CFL-adaptive stability and selectable kinetics;
- Quantitative morphometric and topological analysis (wavelength, Euler characteristic, spacing statistics);
- Calibrated parameter presets for three polypore species;
- Reproducible, contract-enforced output schemas;
- Sensitivity analysis tools for systematic morphospace exploration.

The software targets three user communities: (a) mycologists and developmental biologists studying pattern formation in fungi, (b) computational biologists modeling self-organization, and (c) materials scientists exploring bio-inspired porous architectures.

## 2. Software description

### 2.1. Software architecture

FungalMorphoSpace is organized into five modules:

**core** Turing pattern simulator with polymorphic kinetics models (Gierer–Meinhardt, Schnakenberg, Gray–Scott).

**analysis** Topology analyzer (Euler characteristic, connected components) and morphometric tools (FFT and autocorrelation wavelength, spacing statistics, isotropy index).

**runners** Validation pipeline orchestrating single-species, multi-species, and sensitivity runs with progress reporting and statistical aggregation.

**contracts** Output schema definitions ensuring machine-readable CSV reproducibility across versions.

**utils** Parameter sweep tools and repository cleanup utilities.

Table 1: Code metadata.

| Code metadata description                                      |                                                                                                                  |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Current code version                                           | 0.8.0 (release accompanying this paper)                                                                          |
| Permanent link to code repository                              | <a href="https://github.com/mahumadad/FungalMorphoSpace">https://github.com/mahumadad/<br/>FungalMorphoSpace</a> |
| Permanent link to reproducible capsule                         | <a href="https://doi.org/10.5281/zenodo.20516426">https://doi.org/10.5281/zenodo.20516426</a>                    |
| Legal code license                                             | CC BY-NC 4.0                                                                                                     |
| Code versioning system used                                    | git                                                                                                              |
| Software code languages, tools, services used                  | Python ( $\geq 3.10$ )                                                                                           |
| Compilation requirements, operating environments, dependencies | NumPy, SciPy, scikit-image, pandas, Matplotlib, tqdm, PyYAML                                                     |
| Link to developer documentation                                | README.md and docs/ in the repository                                                                            |
| Support email for questions                                    | <a href="mailto:mahumadad@gmail.com">mahumadad@gmail.com</a>                                                     |

## 2.2. Mathematical model

The core simulator implements the Gierer–Meinhardt system [3]:

$$\begin{aligned}\frac{\partial u}{\partial t} &= D_u \nabla^2 u + \rho \left( \frac{u^2}{v + \varepsilon} - b u + a \right), \\ \frac{\partial v}{\partial t} &= D_v \nabla^2 v + \rho (u^2 - v),\end{aligned}\tag{1}$$

where  $u$  is a short-range activator,  $v$  a long-range inhibitor,  $D_u$  and  $D_v$  are diffusion coefficients,  $\rho$  is the kinetic intensity (a proxy for metabolic activity), and  $\varepsilon = 10^{-6}$  prevents numerical singularity in the autocatalytic term.

The homogeneous steady state is  $u_0 = (a + 1)/b$ ,  $v_0 = u_0^2$ . Classical two-species Turing instability requires a diffusion disparity  $D_v/D_u \gg 1$ , and the pattern wavelength then depends jointly on  $D_v/D_u$  and  $\rho$ . This disparity is not fundamental, however: it is one way to set the underlying reaction–diffusion length  $\ell \sim \sqrt{D_{\text{eff}}/k}$  that fixes the wavelength. An immobile (non-diffusing) third species coupled to the inhibitor produces the same instability at *equal* diffusion, as established for binding substrates and morphogen adsorption [4, 5] and catalogued for three-node networks with equally diffusing signals [6]. FungalMorphoSpace includes an optional three-node module (Sec. 2.3) implementing this mechanism, which lets users contrast disparity-driven and immobile-node routes to the same morphospace.

## 2.3. Three-node (immobile-node) module

The module `core.three_node` adds two diffusing morphogens and a third immobile field  $w$  (interpreted as local hyphal density), with a linear-stability utility (`dispersion_growth_rate`, `fastest_growing_wavelength`)

that computes the Turing band analytically from the reaction Jacobian. It is intended for exploring *whether* a given topology patterns and at what scale *before* committing to a simulation. We stress that the underlying equal-diffusion mechanism is established prior art [4, 5, 6, 7]; the module’s role here is to make that mechanism convenient to apply to hymenophore morphospace, not to introduce a new mechanism.

*Temporal rescaling..* Under the change of variable  $\tau = \rho t$ , the effective diffusion coefficients become  $D_u^* = D_u/\rho$  and  $D_v^* = D_v/\rho$ . This reveals that the characteristic wavelength  $\lambda$  depends on the ratio  $D_v/\rho$ , not  $D_v$  alone—a key insight that resolves cases where increasing  $D_v/D_u$  paradoxically *decreases*  $\lambda$  when  $\rho$  increases simultaneously (the “Brumalis paradox”; see Table 2).

#### 2.4. Numerical implementation

Spatial discretization uses a second-order five-point finite-difference stencil on a square grid ( $N \times N$ ,  $N \in \{256, 512, 1024\}$ ) with periodic boundary conditions. Time integration uses the explicit (forward) Euler method with adaptive time-stepping via the CFL condition:

$$\Delta t \leq \frac{(\Delta x)^2}{4 \max(D_u, D_v)}, \quad (2)$$

reduced by a safety factor of 0.8 to ensure stability for stiff reactions at high  $D_v/D_u$  ratios. Temporal comparability across species is enforced through a target physical time  $T_{\text{target}}$ , from which the number of steps is computed as  $n_{\text{steps}} = \lceil T_{\text{target}}/\Delta t_{\text{final}} \rceil$ .

A critical numerical fix prevents catastrophic blowup in the  $u^2/v$  term: a floor value  $v_{\text{floor}} = 10^{-6}$  is enforced after each time step, complementing the  $\varepsilon$  regularization in the kinetic function.

#### 2.5. Morphometric and topological analysis

Pattern quantification employs two complementary approaches:

*Wavelength estimation..* Two independent methods provide robustness:

1. **FFT radial spectrum:** the 2D power spectrum is radially averaged; the dominant peak above a minimum frequency ( $\geq 2$  cycles per domain) yields  $\lambda_{\text{FFT}}$ . Quality control (QC) requires peak prominence  $\geq 3 \times$  median background.
2. **Autocorrelation:** the 2D autocorrelation (via the Wiener–Khinchin theorem) is radially averaged;  $\lambda_{\text{AC}}$  is the position of the first maximum after the first minimum.

The pipeline selects the best estimate via a QC cascade: FFT (if QC pass)  $\rightarrow$  autocorrelation (if QC pass)  $\rightarrow$  fallback (first finite positive value, marked as QC fail).

80 *Topological metrics..*

- **Euler characteristic**  $\chi = N_{\text{components}} - N_{\text{holes}}$ , computed via `skimage.measure.euler_number` [8] with 8-connectivity. Spots yield  $\chi > 0$ ; labyrinths  $\chi \approx 0$ ; networks  $\chi < 0$ .
- **Connected components**: count and centroids of binarized foreground regions, replacing the noise-sensitive local maxima detection used in earlier versions.
- 85 • **Spacing statistics**: nearest-neighbor distances from component centroids (via KD-tree), yielding mean spacing and coefficient of variation as regularity measures.
- **Isotropy index**: angular distribution of 2D FFT power;  $I = 1$  for perfectly isotropic patterns.

## 2.6. Calibrated species database

Table 2 summarizes the three calibrated species. Parameters were calibrated empirically by adjust-  
 90 ing  $D_v/D_u$ ,  $\rho$ , and  $b$  until simulated  $\lambda$  matched observed pore spacings from the literature. The spatial calibration factor  $8.7 \mu\text{m}/\text{px}$  derives from *F. fomentarius* scanning electron microscopy data [9].

Table 2: Calibrated species parameters and resulting wavelengths.

| Species               | Key              | $D_v/D_u$ | $\rho$ | $b$ | Grid | $\lambda$ (px) | $\lambda$ ( $\mu\text{m}$ ) |
|-----------------------|------------------|-----------|--------|-----|------|----------------|-----------------------------|
| <i>F. fomentarius</i> | <b>fomes</b>     | 150       | 0.2    | 1.0 | 512  | 46             | 400                         |
| <i>L. brumalis</i>    | <b>brumalis</b>  | 250       | 0.4    | 1.0 | 512  | 33             | 261                         |
| <i>P. squamosus</i>   | <b>squamosus</b> | 3750      | 5.0    | 3.0 | 1024 | 235            | 2043                        |

*Transparency note..* These parameters were obtained by empirical fitting, not from independent biophysical measurements. The software therefore serves as an exploration and hypothesis-generation tool rather than an *ab initio* predictor. Strategies for non-tautological parameter anchoring—e.g., deriving  $D_{\text{eff}}$  from measured  
 95 substrate porosity—are discussed in the project documentation.

*Why the presets retain large  $D_v/D_u$ ..* The presets (notably  $D_v/D_u = 3750$  for *P. squamosus*) use the classical two-species parameterization for continuity with the prior literature and for direct comparability across species within one kinetic family. Section 3.5 shows that this large ratio is not the only way to reach the same scale—the diffusion length can equally be set at  $D_u = D_v$  via the immobile-node module—so the  
 100 extreme ratio is a modelling convenience of these presets, not a requirement of the mechanism.

### 3. Illustrative examples

#### 3.1. Multi-species validation

```
1 # Validate all 3 species (5 replicates, high resolution)
105 python scripts/run_integrated_validation.py \
3     --species all --n_runs 5 --grid 1024
```

Listing 1: Full validation suite.

This generates a comprehensive comparison figure with pattern images and wavelength measurements for the three presets, which reproduce the observed pore spacings across an  $8\times$  range. We deliberately report no allometric exponent or goodness-of-fit: with  $N = 3$  calibrated species, any such regression is statistically meaningless and would overstate the evidence. The presets demonstrate *reachability* of the observed scales within one model, not a scaling law.

#### 3.2. Programmatic API

```
115 from fungalmorphospace.core import (
2     TuringSimulator, GiererMeinhardtKinetics)
3 from fungalmorphospace.analysis import TopologyAnalyzer
4
5 # Configure and run
120 kinetics = GiererMeinhardtKinetics(
7     rho=0.2, a=0.1, b=1.0)
8 sim = TuringSimulator(
9     kinetics_model=kinetics,
10     D_u=0.1, D_v=15.0, grid_size=512)
125 sim.initialize(perturbation_amplitude=0.1)
12 sim.run(num_steps=10000)
13
14 # Analyze
15 analyzer = TopologyAnalyzer(sim.u, dx=1.0)
190 metrics = analyzer.compute_all_metrics()
17 print(f"Lambda: {metrics['wavelength_fft']:.1f} px")
18 print(f"Euler: {metrics['euler_characteristic']}")
```

Listing 2: Python API usage.

### 3.3. Topological continuity

The script `test_laminillas.py` demonstrates that pores and gills are topological variants: varying  $D_v/D_u$  while tracking  $\chi$  reveals a smooth transition from isolated spots ( $\chi > 0$ ) through labyrinths ( $\chi \approx 0$ ) to connected networks ( $\chi < 0$ ), all generated by the same Gierer–Meinhardt mechanism.

### 3.4. Sensitivity analysis

```
140 from fungalmorphospace.utils.sensitivity_analysis \
2     import SensitivityAnalyzer
3
4 sa = SensitivityAnalyzer(base_params={
5     'D_u': 1.0, 'grid_size': 256,
145     'dx': 1.0, 'dt': 0.0005,
7     'steps': 5000, 'model': 'gierer_meinhardt'})
8
9 df = sa.sweep_2d(
10     'D_v_D_u', [50, 100, 200, 400],
150     'rho', [0.1, 0.2, 0.5, 1.0])
```

Listing 3: Parameter sweep.

### 3.5. Equal-diffusion exploration and the wavelength scale law

The three-node module (Sec. 2.3) lets users probe whether the hymenophore morphospace can be reached *without* a diffusion disparity. Using the analytical dispersion utility, a topology in which the immobile node couples to the inhibitor is Turing-unstable at  $D_u = D_v$ , and its fastest-growing wavelength follows the reaction–diffusion length scaling  $\lambda \propto \sqrt{D}$  (Fig. 1). Nonlinear simulations reproduce this scaling: the measured spot spacing tracks  $\sqrt{D}$  with a constant offset above the linear  $\lambda^*$  (coarsening factor  $\approx 2.4$ , independent of  $D$  and of the saturation strength). The scripts `search_turing_topology.py`, `confirm_three_node.py`, and `figure_scale_law.py` reproduce these results.

This example illustrates the software’s analytical value: it shows that the extreme ratio  $D_v/D_u = 3750$  used for the coarse-pored *P. squamosus* preset (Table 2) is not a necessary feature of the *model* but one parameterization of the diffusion length; the same scale is reachable at equal diffusion via the immobile node. We stress that this is a statement about model parameterization and known mechanism classes, *not* a biological claim: it does not show that extreme differential diffusion is absent in fungi, nor that the immobile node has a confirmed fungal correlate. The interpretation of  $w$  as local hyphal density is a hypothesis; discriminating among these mechanisms in real polypores requires the measurements discussed in Sec. 4.

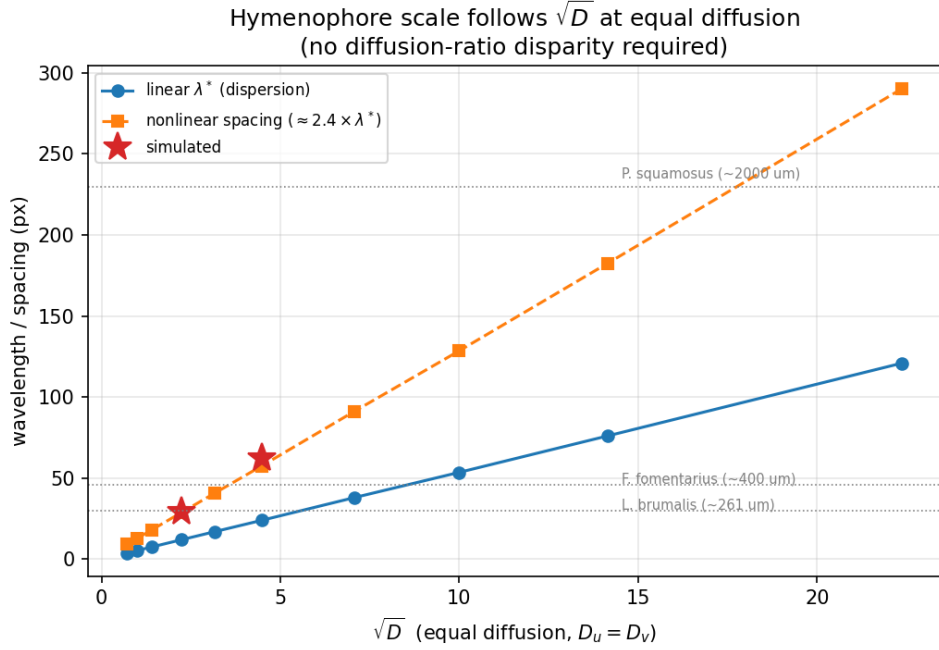


Figure 1: Wavelength versus  $\sqrt{D}$  at equal diffusion ( $D_u = D_v$ ) for the immobile-node topology. Linear dispersion prediction (circles) and nonlinear simulated spacing (squares) are both straight lines in  $\sqrt{D}$ ; simulated points (stars) lie on the nonlinear curve. Dotted lines mark the biological pore spacings of the three calibrated species. Generated by `figure_scale_law.py`.

## 4. Impact

### 4.1. Scientific contributions

FungalMorphoSpace has supported:

- Computational *exploration* of the Kuhar hypothesis [1]: the three calibrated species, spanning an  $8\times$  wavelength range, are reproducible as parametric variants of one Gierer–Meinhardt mechanism (consistency, not independent validation; see Limitations);
- A transparent treatment of the  $D/\rho$  nondimensionalization (effective diffusion), which clarifies cases where increasing  $D_v/D_u$  while raising  $\rho$  *decreases*  $\lambda$ ;
- Demonstration of topological continuity between morphotypes (spots  $\leftrightarrow$  labyrinths) within a single model;
- An analytical demonstration that the morphospace scale is governed by a diffusion length and is reachable at equal diffusion via an immobile node (Sec. 3.5), so that extreme diffusion ratios are not a necessary model ingredient.



#### 4.2. Broader applications

Beyond mycology, the software provides:

- A pedagogical tool for teaching pattern formation and self-organization in undergraduate or graduate courses;
- A generative engine for bio-inspired porous architectures in biomaterials design;
- A baseline framework for integrating molecular data with morphogenetic models.

#### 4.3. Limitations and future directions

##### Current limitations:

- *Empirical calibration (the central caveat).* Parameters are fitted to observed  $\lambda$ , not derived from independent biophysical measurements, so reproducing a wavelength is a consistency check, not a prediction. Because  $\lambda$  is set by a single effective reaction–diffusion length, many parameter combinations and indeed several distinct mechanisms reproduce the same  $\lambda$ ; breaking this degeneracy requires measuring the underlying diffusion and reaction rates (Future directions).
- *The pattern-forming mechanism is not novel.* Equal-diffusion Turing via an immobile/binding species is established theory, including applications to other organisms [4, 5, 6, 7]; the contribution here is the application to fungal hymenophores, the accompanying reproducible tooling, and the demonstration that the prior extreme-ratio parameterization is unnecessary.
- *Stability caveat for the immobile-node module.* Reaction-diffusion-ODE systems with an autocatalytic immobile component need not admit stable Turing patterns [10]; the preset topology used here is non-autocatalytic (binding-type), and the nonlinear checks reported here are finite-time, so long-time stability of the patterns should be verified per application.
- Simulations are 2D; real hymenophores are 3D surfaces (locally planar approximations are reasonable).
- Only  $N = 3$  species are calibrated, giving limited statistical power for scaling-law inference.
- The explicit Euler solver, while stable with CFL control, becomes expensive at very high  $D_v/D_u$  ratios and large grids (e.g. *P. squamosus*-scale wavelengths).

##### Planned extensions:

- Non-tautological parameter anchoring via independently measured diffusion and reaction rates—e.g. substrate porosity and microstructure [9] and  $\text{Ca}^{2+}$  signal-propagation data in mycelial networks [11]—which would discriminate the candidate mechanisms rather than fit the wavelength;

- Extension to  $N > 10$  species for robust allometric scaling;
- 3D simulations with surface curvature effects;
- Machine learning for parameter inference from microscopy images;
- Implicit or semi-implicit solvers for improved efficiency.

## 5. Conclusions

FungalMorphoSpace shows that a single Gierer–Meinhardt parameterization *can be fitted* to reproduce the pore spacings of three polypore species across an  $8\times$  range, and that the same morphospace is also reachable by an established equal-diffusion mechanism. We are careful to distinguish this—reachability of the observed scales within a model class—from the much stronger and unsupported claim that fungi *use* this mechanism: the parameters are fitted to the data they reproduce, so the result is a consistency check, not a demonstration of a developmental mechanism. The software’s value is therefore as a reproducible, well-documented pipeline for simulating patterns, quantifying morphology, and generating falsifiable hypotheses—hypotheses whose test requires the independent biophysical measurements outlined in Sec. 4, not further fitting.

The package is available under a CC BY-NC 4.0 license at <https://github.com/mahumadad/FungalMorphoSpace>.

## Code and data availability

The source code is available under a CC BY-NC 4.0 license at <https://github.com/mahumadad/FungalMorphoSpace>. The versioned release (v0.8.0) accompanying this paper is archived on Zenodo at <https://doi.org/10.5281/zenodo.20516426>. All figures and tables in this paper are reproducible with the scripts under `scripts/` and the configuration in `data/species_data.json`.

## CRediT author contributions

**Mario Ahumada Durán:** Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing.

## Declaration of generative AI use

The author used an AI assistant (Claude) to support code review, literature search and verification, and drafting. All AI-assisted outputs—including every cited reference—were verified by the author, who takes full responsibility for the content.

## Funding

This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## Conflict of interest

The author declares no competing financial or non-financial interests.

## Acknowledgements

The author thanks the developers of the open-source scientific Python ecosystem (NumPy, SciPy, scikit-image, Matplotlib) on which this software is built.

## References

- [1] F. Kuhar, L. Terzzoli, E. Nouhra, G. Robledo, M. Mercker, Pattern formation features might explain homoplasia: fertile surfaces in higher fungi as an example, *Theory in Biosciences* 141 (1) (2022) 1–11. doi:10.1007/s12064-022-00363-z.
- [2] A. M. Turing, The chemical basis of morphogenesis, *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* 237 (641) (1952) 37–72. doi:10.1098/rstb.1952.0012.
- [3] A. Gierer, H. Meinhardt, A theory of biological pattern formation, *Kybernetik* 12 (1) (1972) 30–39. doi:10.1007/BF00289234.
- [4] K. Korvasová, E. A. Gaffney, P. K. Maini, M. A. Ferreira, V. Klika, Investigating the Turing conditions for diffusion-driven instability in the presence of a binding immobile substrate, *Journal of Theoretical Biology* 367 (2015) 286–295. doi:10.1016/j.jtbi.2014.11.024.
- [5] A. M. Nesterenko, M. B. Kuznetsov, D. D. Korotkova, A. G. Zaráisky, Morphogene adsorption as a Turing instability regulator: Theoretical analysis and possible applications in multicellular embryonic systems, *PLOS ONE* 12 (2) (2017) e0171212. doi:10.1371/journal.pone.0171212.
- [6] L. Marcon, X. Diego, J. Sharpe, P. Müller, High-throughput mathematical analysis identifies Turing networks for patterning with equally diffusing signals, *eLife* 5 (2016) e14022. doi:10.7554/eLife.14022.
- [7] J. Raspopovic, L. Marcon, L. Russo, J. Sharpe, Digit patterning is controlled by a Bmp-Sox9-Wnt Turing network modulated by morphogen gradients, *Science* 345 (6196) (2014) 566–570. doi:10.1126/science.1252960.
- [8] S. van der Walt, J. L. Schönberger, J. Nunez-Iglesias, F. Boulogne, J. D. Warner, N. Yager, E. Gouillart, T. Yu, scikit-image: image processing in Python, *PeerJ* 2 (2014) e453. doi:10.7717/peerj.453.
- [9] S. Klemm, L. Kerschbaumer, J. Tintner, N. Gierlinger, W. Gindl-Altmutter, M. Meng, J. Konnerth, Hierarchical structure, mechanical properties, and chemical composition of *Fomes fomentarius* across the stem, the context, and the hymenophore, *PLOS ONE* 19 (5) (2024) e0303122. doi:10.1371/journal.pone.0303122.
- [10] A. Marciniak-Czochra, G. Karch, K. Suzuki, Instability of Turing patterns in reaction-diffusion-ODE systems, *Journal of Mathematical Biology* 74 (2016) 583–618. doi:10.1007/s00285-016-1035-z.
- [11] A. Itani, S. Masuo, R. Yamamoto, T. Serizawa, Y. Fukasawa, N. Takaya, M. Toyota, S. Betsuyaku, N. Takeshita, Local calcium signal transmission in mycelial network exhibits decentralized stress responses, *PNAS Nexus* 2 (3) (2023) pgad012. doi:10.1093/pnasnexus/pgad012.